

Gaudin Hamiltonians from the collision residue: the seven-face identification at genus zero

Raez Lorgat

Abstract

For a chirally Koszul chiral algebra $\mathcal{A}$ with universal Maurer–Cartan element $\Theta_{\mathcal{A}}$, the collision residue $r(z) = \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}})$ encodes the genus-zero binary shadow. The Gaiotto–Zeng commuting Hamiltonians $\{H_i\}$ on $\mathcal{M}_{0,n}$ are the components of the flat connection $\nabla = d - \sum_{i < j} r(z_{ij}) dz_{ij}$ built from $r(z)$. For affine Kac–Moody algebras $\widehat{\mathfrak{g}}_k$, these Hamiltonians are the Gaudin Hamiltonians $H_i^{\text{Gaudin}} = \sum_{j \neq i} \Omega_{ij} / (z_i - z_j)$ of Feigin–Frenkel–Reshetikhin, rescaled by $1/(k + h^\vee)$. For a general algebra $\mathcal{A}$ with maximal OPE pole order $p_{\max}$, the collision depth $k_{\max} = p_{\max} - 1$ determines the operator order: the Hamiltonians are multiplication operators for $k_{\max} \leq 1$, trivial for $k_{\max} = 0$, and differential operators of order $k_{\max} - 1$ for $k_{\max} \geq 3$. This operator-order trichotomy partitions the standard landscape into three regimes. For $\mathcal{W}_3$, the prediction is 4th-order differential operators from $k_{\max} = 5$; the computational checks are recorded separately in Section 7.

I Introduction

The Gaudin model, introduced by Gaudin in 1976 and placed in the framework of quantum groups by Feigin, Frenkel, and Reshetikhin [1], assigns to n distinct points $z_1, \dots, z_n$ on $\mathbb{P}^1$ a commutative algebra of Hamiltonians acting on $V_{\lambda_1} \otimes \dots \otimes V_{\lambda_n}$, where the V_{λ_i} are $\mathfrak{g}$ -modules. The Gaudin Hamiltonians are

$$H_i^{\text{Gaudin}} = \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\Omega_{ij}}{z_i - z_j}, \quad i = 1, \dots, n, \quad (1.1)$$

where $\Omega_{ij} = \sum_a t_i^a \otimes (t_a)_j$ is the Casimir tensor acting in the i -th and j -th factors.

In the 2026 interface-holography paper of Gaiotto–Zeng [2], the sphere Hamiltonians attached to chiral OPE data form a commuting family. On the affine Kac–Moody comparison surface, the corresponding kernel is the KZ-normalized collision residue, so

$$H_i^{\text{GZ}} = \frac{1}{k + h^\vee} H_i^{\text{Gaudin}}. \quad (1.2)$$

For a general algebra in the standard bosonic chiral Koszul landscape, the GZ Hamiltonians are richer: the collision residue may have poles of order up to $k_{\max} = p_{\max} - 1$ (the collision depth), and each pole contributes a term of increasing differential operator order after the conformal-block or Ward-identity reduction. The result is a family of commuting differential operators whose order is controlled by $k_{\max}$ inside this standard landscape.

The affine comparison gives (1.2); the pole filtration gives the operator-order trichotomy; the same calculation applies to the class- $\mathcal{M}$ Virasoro and $\mathcal{W}_N$ examples after the Ward-identity or $\mathcal{W}$ -constraint reduction appropriate to the conformal-block model.

2 The collision residue

2.1 Definition and pole structure

Definition 2.1 (Collision residue). Let $\mathcal{A}$ be a chirally Koszul chiral algebra with universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \text{MC}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$. The *collision residue* is the genus-zero binary projection

$$r(z) := \text{Res}_{0,2}^{\text{coll}}(\Theta_{\mathcal{A}}) \in \mathcal{A}^! \otimes \mathcal{A}^![[z^{-1}]].$$

It admits a Laurent expansion $r(z) = \sum_{m=1}^{k_{\max}} r_m z^{-m}$ with $r_m \in \mathcal{A}^! \otimes \mathcal{A}^!$.

The collision residue lives on the ordered bar complex $B^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct; it is an element of the E_1 convolution algebra $\mathfrak{g}_{\mathcal{A}}^{E_1}$ that retains the full spectral-parameter dependence. The Gaudin Hamiltonians are constructed from this ordered datum; the modular characteristic κ is the Σ_2 -coinvariant scalar shadow, given by $\text{av}(r(z))$ in abelian and scalar families and by $\text{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody.

Proposition 2.2 (Pole structure). *The collision residue $r(z)$ has poles of order at most $k_{\max} = p_{\max} - 1$, where $p_{\max}$ is the maximal OPE pole order among generating fields. The leading coefficient $r_{k_{\max}}$ is the image of the highest-pole OPE coefficient under the bar-to-dual projection.*

Proof. The bar propagator $d \log(z - w) = dz/(z - w)$ absorbs one power of the singularity: a pole of order n in the OPE $a(z)b(w) \sim c \cdot (z - w)^{-n}$ produces a residue of order $n - 1$ in the collision expansion. The maximal OPE pole is $p_{\max}$, so the maximal collision pole is $p_{\max} - 1 = k_{\max}$. $\square$

2.2 The affine Kac–Moody case

For $\widehat{\mathfrak{g}}_k$ with generators $\{J^a\}$ of conformal weight 1, the OPE is

$$J^a(z)J^b(w) \sim \frac{k(t^a, t^b)}{(z - w)^2} + \frac{f^{ab}_c J^c(w)}{z - w}. \quad (2.1)$$

The maximal OPE pole order is $p_{\max} = 2$, so $k_{\max} = 1$ and the collision residue has a single simple pole. The identification of this simple pole with the Feigin–Frenkel–Reshetikhin r -matrix is the genuine content of this section; we isolate it as Theorem 2.3 below.

Theorem 2.3 (KZ-normalized collision residue and FFR r -matrix). *Let $\widehat{\mathfrak{g}}_k$ be the affine Kac–Moody algebra at non-critical level $k \neq -h^\vee$, with $\mathfrak{g}$ simple finite-dimensional and $h^\vee$ the dual Coxeter number. The trace-form collision residue and the KZ-normalized Gaudin kernel are*

$$r_{\text{tr}}^{\text{KM}}(z) = k \Omega_{\text{tr}}/z, \quad r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k + h^\vee)z), \quad \Omega_{\text{KZ}} = 2h^\vee \Omega_{\text{tr}}.$$

Thus the trace-form collision residue extracted from $\Theta_{\widehat{\mathfrak{g}}_k}$ is $k\Omega_{\text{tr}}/z$, while the Feigin–Frenkel–Reshetikhin Gaudin kernel is the KZ-normalized connection coefficient:

$$r_{\text{Gaudin}}(z) := \frac{\Omega}{(k + h^\vee)z}, \quad (2.2)$$

with $\Omega = \sum_a t^a \otimes t_a$ the quadratic Casimir. The trace-form and KZ-normalized expressions are conventionally related normalizations, not identical rational functions of k .

Proof. (a) **Simple-pole support from the OPE.** By Proposition 2.2, $k_{\text{max}} = p_{\text{max}} - 1 = 1$, hence the Laurent expansion reduces to $r(z) = r_1/z$ with $r_1 \in \widehat{\mathfrak{g}}_k^\dagger \otimes \widehat{\mathfrak{g}}_k^\dagger$ the image under the bar-to-dual projection of the leading OPE coefficient of (2.1). The double-pole coefficient $k(t^a, t^b)$ gives the trace-form central tensor $k\Omega_{\text{tr}}/z$. The structure-constant simple pole gives the Lie bracket part of the current algebra; it is not contracted with the central tensor into a second copy of Ω .

(b) **KZ normalisation fixes the $(k + h^\vee)^{-1}$ connection coefficient.** The KZ connection is constructed through the Sugawara stress tensor $T(z) = \frac{1}{2(k+h^\vee)} : J^a J_a : (z)$, whose normalisation factor $(k + h^\vee)^{-1}$ enters via the identity $[T, J^a] = \partial J^a$, which holds at non-critical level $k \neq -h^\vee$ and fails at $k = -h^\vee$ precisely because the Sugawara vector is not defined in the affine vacuum module. Passing from the trace-form collision tensor to the KZ connection therefore replaces $k\Omega_{\text{tr}}$ by $\Omega_{\text{KZ}}/(k + h^\vee)$. This is a normalization bridge, not an algebraic contraction of the central and Lie OPE terms. In KZ normalization, $r_1 = \Omega/(k + h^\vee)$, which is (2.2).

(c) **Trace-form and KZ normalizations.** The Drinfeld Yangian classical kernel is $r_Y(z) = \Omega_{\text{tr}}/z$. For $\mathfrak{g}$ simple, $\Omega_{\text{tr}} = (2h^\vee)^{-1}\Omega$ after passing from the trace form to the normalized invariant form [8, §6]. Thus $k\Omega_{\text{tr}}/z$ and $\Omega/((k + h^\vee)z)$ are not identical rational functions of k ; they agree only after the normalization bridge fixed by the Sugawara/KZ convention. This is the comparison identifying the Yangian and Gaudin faces in the affine genus-zero kernel. $\square$

Equation (2.2) rewritten with explicit scalar gives the form used elsewhere in the paper:

$$r(z) = \frac{\Omega}{(k + h^\vee)z}.$$

The numerator $\Omega = \sum_a t^a \otimes t_a$ is the quadratic Casimir, and the denominator $(k + h^\vee)$ is the Sugawara normalization.

Corollary 2.4 (Critical-level Feigin–Frenkel limit). *The Gaudin Hamiltonians $H_i^{\text{Gaudin}} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$ at critical level $k = -h^\vee$ arise from the finite critical limit of the rescaled Segal–Sugawara fields. The unrescaled KZ connection is singular at $k = -h^\vee$; the Feigin–Frenkel centre supplies the critical Gaudin algebra, and its quadratic generators have the classical shadow H_i^{Gaudin} .*

Proof. For $k \neq -h^\vee$, the KZ kernel is $\Omega/((k + h^\vee)z)$. Multiplying the Segal–Sugawara fields by $k + h^\vee$ removes the pole in the Sugawara normalization and has a finite critical limit. Feigin–Frenkel identify these limiting fields with generators of the centre at the critical level; Feigin–Frenkel–Reshetikhin identify their images with the Gaudin Hamiltonians [1, 11]. Thus the critical Gaudin algebra is the central limit of the rescaled Sugawara construction, not the value of the unrescaled KZ connection at $k = -h^\vee$. $\square$

3 The Gaudin identification

Theorem 3.1 (Gaudin from collision residue). *For $\widehat{\mathfrak{g}}_k$ at non-critical level ($k \neq -h^\vee$), the GZ Hamiltonians on $\mathcal{M}_{0,n}$ are the Gaudin Hamiltonians rescaled by $1/(k + h^\vee)$:*

$$H_i^{\text{GZ}} = \frac{1}{k + h^\vee} H_i^{\text{Gaudin}} = \frac{1}{k + h^\vee} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{\Omega_{ij}}{z_i - z_j}.$$

In particular, $[H_i^{\text{GZ}}, H_j^{\text{GZ}}] = 0$ for all i, j .

Proof. The GZ Hamiltonians are defined as the residues of the flat connection

$$\nabla^{\text{GZ}} = d - \sum_{i < j} r(z_i - z_j) dz_{ij} \quad (3.1)$$

on $\mathcal{M}_{0,n}$, where $r(z)$ is the collision residue of $\Theta_{\mathcal{A}}$. Flatness of ∇^{GZ} is equivalent to the Maurer–Cartan equation: $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at genus zero projects to $[H_i, H_j] = 0$.

For $\widehat{\mathfrak{g}}_k$, substituting $r(z) = \Omega/((k + h^\vee)z)$ (Theorem 2.3) into (3.1) gives

$$\nabla^{\text{GZ}} = d - \frac{1}{k + h^\vee} \sum_{i < j} \frac{\Omega_{ij}}{z_i - z_j} dz_{ij}.$$

Taking the coefficient of dz_i gives

$$H_i^{\text{GZ}} = \frac{1}{k + h^\vee} \sum_{j \neq i} \frac{\Omega_{ij}}{z_i - z_j} = \frac{H_i^{\text{Gaudin}}}{k + h^\vee}. \quad \square$$

Remark 3.2 (Commutativity from MC). The commutativity $[H_i^{\text{Gaudin}}, H_j^{\text{Gaudin}}] = 0$ is a classical result proved by Feigin–Frenkel–Reshetikhin [1] using the Bethe ansatz. In the present framework, it is the genus-zero binary consequence of the Maurer–Cartan equation. The projection of $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ is the infinitesimal braid relation

$$[\Omega_{ij}, \Omega_{kl}] = 0, \quad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0$$

for distinct indices. These identities imply $[H_i^{\text{GZ}}, H_j^{\text{GZ}}] = 0$.

Remark 3.3 (The KZ connection). The Knizhnik–Zamolodchikov connection $\nabla^{\text{KZ}} = d - (1/(k + h^\vee)) \sum_{i < j} \Omega_{ij} d \log(z_i - z_j)$ on the space of conformal blocks is the original example. The GZ connection ∇^{GZ} generalizes it from affine Kac–Moody to arbitrary chiral algebras: for non-affine algebras, the connection acquires higher-order terms corresponding to higher-collision-depth poles. For affine Kac–Moody, the GZ and KZ connections coincide.

4 The operator-order trichotomy

The collision depth $k_{\max}$ determines the differential-operator order of the GZ Hamiltonians. Three regimes arise.

Theorem 4.1 (Operator-order trichotomy in the standard landscape). *Let $\mathcal{A}$ belong to the standard bosonic chiral Koszul landscape and let $k_{\max} = p_{\max} - 1$ be its collision depth. After the conformal-block or Ward-identity reduction, the GZ Hamiltonians H_i^{GZ} on $\mathcal{M}_{0,n}$ are:*

- (i) Trivial ($H_i = 0$) if $k_{\max} = 0$, i.e., $p_{\max} = 1$.
- (ii) Multiplication operators (order 0) if $k_{\max} = 1$, i.e., $p_{\max} = 2$.
- (iii) Differential operators of order $k_{\max} - 1$ if $k_{\max} \geq 3$, i.e., $p_{\max} \geq 4$.

Proof. The collision residue $r(z) = \sum_{m=1}^{k_{\max}} r_m z^{-m}$ has poles up to order $k_{\max}$. The GZ connection ∇^{GZ} is built from $r(z)$ evaluated at $z = z_i - z_j$: each pole of order m in $r(z)$ contributes a term $(z_i - z_j)^{-m}$ to the connection. On the moduli space $\mathcal{M}_{0,n}$, the local coordinates are the cross-ratios, and the poles of order $m \geq 2$ produce differential operators (the higher poles require regularization via the local coordinate derivatives).

Concretely:

- If $k_{\max} = 0$: the collision residue vanishes identically (the bar propagator absorbs the only pole), so $H_i = 0$. This is the first-order free-field regime, including the standard $\beta\gamma$ and free-fermion examples.
- If $k_{\max} = 1$: the collision residue is $r(z) = r_1/z$ (simple pole only), and $H_i = \sum_{j \neq i} r_1^{ij} / (z_i - z_j)$ is a multiplication operator on $\oplus V_{\lambda_i}$. This is the affine Kac–Moody regime: the Gaudin Hamiltonians are multiplication operators (they act on the tensor product of $\mathfrak{g}$ -modules, not on functions of the z_i).
- If $k_{\max} \geq 3$: the collision residue has poles $z^{-1}, z^{-2}, \dots, z^{-k_{\max}}$, and the higher poles produce differential operators. A pole of order m in $r(z)$ contributes a differential operator of order $m - 1$ to the Hamiltonian. The maximal order is $k_{\max} - 1$. This is the Virasoro/ $\mathcal{W}_N$ regime.

□

Remark 4.2 (Why $k_{\max} = 2$ is absent). The value $k_{\max} = 2$ does not occur in the standard landscape. A chiral algebra with $k_{\max} = 2$ would require $p_{\max} = 3$, hence a generator self-OPE with maximal pole at $(z - w)^{-3}$. Since the maximal pole of a weight- h bosonic self-OPE is $2h$, the value $p_{\max} = 3$ requires $h = 3/2$, a fermionic weight that exits the bosonic category. The trichotomy is therefore $k_{\max} \in \{0, 1\} \cup \{3, 4, 5, \dots\}$.

Remark 4.3 (Why $k_{\max} - 1$, not $k_{\max}$). The operator order is $k_{\max} - 1$, not $k_{\max}$, because the simple-pole term r_1/z always contributes a multiplication operator (order 0). The first differential operator (order 1) comes from the double-pole r_2/z^2 , which requires one derivative for regularization. In general, the m -th pole contributes order $m - 1$.

The trichotomy partitions the standard landscape:

Regime	$k_{\max}$	Operator order	Families
Trivial	0	$H_i = 0$	$\beta\gamma$, free fermion
Gaudin	1	0 (multiplication)	affine $\widehat{\mathfrak{g}}_k$, Heisenberg
Differential	≥ 3	$k_{\max} - 1$	Virasoro ($k_{\max} = 3$), $\mathcal{W}_N$ ($k_{\max} = 2N - 1$)

Remark 4.4 (Shadow depth and operator order). The shadow depth $r_{\max}$ does not determine the operator order. The $\beta\gamma$ system has $r_{\max} = 4$ (class C) but $k_{\max} = 0$ (trivial Hamiltonians). The Heisenberg

algebra has $r_{\max} = 2$ (class G) and $k_{\max} = 1$ (Gaudin regime). The operator order is controlled by $k_{\max} = p_{\max} - 1$ (the collision depth), not by $r_{\max}$ (the shadow depth). The two invariants are independent (Proposition 2.4 of the classification trichotomy paper).

5 Higher collision depths

5.1 The Virasoro algebra: $k_{\max} = 3$

The Virasoro algebra has generators T of weight 2 with OPE

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The maximal OPE pole is $p_{\max} = 4$, so $k_{\max} = 3$. The collision residue after d log absorption is

$$r_{\text{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z} \in \text{Vir}^1 \otimes \text{Vir}^1 [[z^{-1}]].$$

The even-order poles (z^{-2} , z^{-4}) of the OPE become odd-order poles (z^{-1} , z^{-3}) in the collision residue; the odd-order pole (z^{-1}) of the OPE does not appear in the collision residue because d log absorption sends it to z^0 (a regular term). The result is that $r_{\text{Vir}}(z)$ has only odd-order poles: z^{-3} and z^{-1} .

Proposition 5.1 (Virasoro GZ Hamiltonians). *The Virasoro collision residue gives second-order GZ operators after the Ward-identity reduction on conformal blocks. On primary insertions of conformal weights h_j , the local Ward form is*

$$H_i^{\text{GZ}} = \sum_{j \neq i} \left[\frac{h_j}{(z_i - z_j)^2} + \frac{\partial_{z_j}}{z_i - z_j} \right],$$

with the quadratic stress-tensor channel supplying the second-order symbol in the full Virasoro GZ operator. The collision-depth bound is $\text{ord}(H_i^{\text{GZ}}) \leq k_{\max} - 1 = 2$, and the central z^{-3} pole is the source of the top symbol.

Remark 5.2 (Parity of collision poles). For a bosonic algebra, the d log absorption sends z^{-2n} to $z^{-(2n-1)}$ (even to odd). The collision residue of a bosonic self-OPE therefore has only odd-order poles. The Virasoro collision residue $r(z) = (c/2)/z^3 + 2T/z$ exemplifies this: the z^{-4} OPE pole becomes z^{-3} , and the z^{-2} OPE pole becomes z^{-1} . The z^{-1} OPE pole (the ∂T term) becomes z^0 and does not contribute to the Laurent principal part.

5.2 The $\mathcal{W}_3$ prediction: 4th-order operators

The $\mathcal{W}_3$ algebra has generators T (weight 2) and W (weight 3). The W -self-OPE has maximal pole $p_{\max}(WW) = 6$, so $k_{\max}(\mathcal{W}_3) = 5$. The central and primary collision channels have odd principal poles z^{-5} , z^{-3} , z^{-1} . Lower-symbol terms coming from composite fields and the W -constraint reduction can also contribute even intermediate powers such as z^{-2} ; these do not change the top-order bound.

Proposition 5.3 ($\mathcal{W}_3$ GZ Hamiltonians). *The GZ Hamiltonians for the $\mathcal{W}_3$ algebra are 4th-order differential operators on $\mathcal{M}_{0,n}$. The operator order is $k_{\max} - 1 = 4$. The leading z^{-5} pole produces the*

4th-order term; the z^{-3} pole produces the 2nd-order term (matching the Virasoro contribution from the T -sector); the z^{-1} pole produces the multiplication term.

Proof. The W -self-OPE at pole order 6 gives a collision-residue pole of order 5. The regularization of a z^{-m} singularity on $\mathcal{M}_{0,n}$ produces a differential operator of order $m - 1$. At $m = 5$: order 4. Commutativity $[H_i, H_j] = 0$ follows from the MC equation. $\square$

5.3 The general $\mathcal{W}_N$ pattern

For the principal $\mathcal{W}_N$ algebra ($N \geq 2$), the highest-weight generator W_N has conformal weight N and self-OPE with maximal pole $p_{\max}(W_N W_N) = 2N$. Therefore $k_{\max}(\mathcal{W}_N) = 2N - 1$ and the GZ Hamiltonians are differential operators of order $2N - 2$:

Algebra	$p_{\max}$	$k_{\max}$	Operator order	Shadow class
$\widehat{\mathfrak{g}}_k$	2	1	0	L
Virasoro ($\mathcal{W}_2$)	4	3	2	$\mathcal{M}$
$\mathcal{W}_3$	6	5	4	$\mathcal{M}$
$\mathcal{W}_4$	8	7	6	$\mathcal{M}$
$\mathcal{W}_N$	$2N$	$2N - 1$	$2N - 2$	$\mathcal{M}$

Remark 5.4 (Higher Gaudin from higher collision depth). For $k_{\max} \geq 3$, the collision residue encodes “higher Gaudin Hamiltonians”: the standard Gaudin Hamiltonians (1.1) correspond to the $k = 1$ (simple-pole) truncation. The higher poles produce operators that commute with the Gaudin Hamiltonians and with each other, extending the integrable system from the Gaudin model to a larger commutative algebra. For $\mathcal{W}_N$, the generators of weights $2, \dots, N$ produce the corresponding hierarchy of collision-depth symbols, with top operator order $2N - 2$.

6 The seven-face identification at genus zero

The Gaudin identification is one of the seven faces of the collision residue. The full seven-face chain at genus zero is:

- (F1) *Twisting morphism*: $r(z)$ is the genus-zero binary projection of $\pi_{\mathcal{A}}: B(\mathcal{A}) \rightarrow \mathcal{A}$.
- (F2) *Line-operator R -twist*: $r(z)$ governs the meromorphic tensor product (Dimofte–Niu–Py [3]).
- (F3) *Classical PVA r -matrix*: $r(z)$ is the classical r -matrix of $\text{PVA}(\mathcal{A})$ (Khan–Zeng [4]).
- (F4) *GZ commuting differentials*: $r(z)$ generates the commuting Hamiltonians $\{H_i\}$ (Gaiotto–Zeng [2]).
- (F5) *Yangian R -matrix*: $r(z)$ is the classical limit of $R(z; \hbar)$ with $\hbar = 1/(k + h^\vee)$ (Drinfeld [5]).
- (F6) *Sklyanin bracket*: $r(z)$ generates the Poisson bracket (Semenov-Tian-Shansky [9]).
- (F7) *Gaudin generator*: $r(z)$ generates the Gaudin Hamiltonians (Feigin–Frenkel–Reshetikhin [1]).

Theorem 3.1 establishes $F_4 \Leftrightarrow F_7$ for affine Kac–Moody. The remaining links are:

Proposition 6.1 (The chain at genus zero). *For affine Kac–Moody at non-critical level:*

- (i) $F_1 \Leftrightarrow F_7$: the twisting morphism at genus zero gives the trace-form collision kernel $k\Omega_{\text{tr}}/z$; after the Sugawara/KZ normalization bridge this becomes $\Omega/((k + h^\vee)z)$, whose multi-point extension is the Gaudin Hamiltonian kernel.
- (ii) $F_5 \Leftrightarrow F_6$: the Yangian quantizes the Sklyanin bracket (Drinfeld 1985, Semenov-Tian-Shansky 1983).
- (iii) $F_4 \Leftrightarrow F_7$: Theorem 3.1.
- (iv) $F_1 \Leftrightarrow F_3$: the PVA λ -bracket at leading order is the collision residue (the three-parameter identification).
- (v) $F_1 \Leftrightarrow F_2$: the bar-cobar R -matrix is the line-operator R -twist (DNP identification).

These comparisons identify the same genus-zero binary kernel under the stated normalizations. They do not identify the ambient source objects: the ordered bar twisting morphism, line-operator tensor category, PVA bracket, GZ connection, Yangian R -matrix, Sklyanin bracket, and Gaudin algebra remain distinct structures.

6.1 Elliptic Gaudin, Belavin–Drinfeld, and Felder

At genus one, KZ regularizes to the KZB connection [12]. The non-dynamical elliptic classical r -matrix is the Belavin–Drinfeld matrix [13, 14], with the full Weierstrass ζ_τ in the Cartan channel and theta-function ratios in the root channels. The Felder $R(z, \lambda, \tau)$ is a distinct dynamical IRF/face-model object satisfying the dynamical Yang–Baxter equation [15–17]; for $\mathfrak{sl}_2$ it is related to the Belavin matrix by a vertex–IRF gauge transform, not identical to the bare collision residue.

Theorem 6.2 (Elliptic Gaudin from chiral collision at $g = 1$). *Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ and $\widehat{\mathfrak{sl}}_2$ at non-critical level k . The genus-one collision residue of $\Theta_{\widehat{\mathfrak{sl}}_2}^{\times n}/\Sigma_n$ produces the KZB/Belavin elliptic kernel. After the vertex–IRF gauge transform, the corresponding dynamical object is the Felder R -matrix*

$$R_{\text{Felder}}(z, \lambda, \tau) = \text{id} + \hbar \cdot \frac{\theta'_1(z, \tau)}{\theta_1(z, \tau)} \cdot \Omega(\lambda) + O(\hbar^2),$$

where $\Omega(\lambda) = \Omega_{\mathfrak{sl}_2} + \lambda \cdot h \otimes h$ is the dynamical Casimir, and the elliptic Gaudin Hamiltonians are

$$H_i^{\text{ell}}(\tau, \lambda) = \sum_{j \neq i} \frac{\theta'_1(z_i - z_j, \tau)}{\theta_1(z_i - z_j, \tau)} \Omega_{ij}(\lambda) + \text{KZB terms}.$$

Flatness of the KZB connection $\nabla_{\text{KZB}} = d - \sum_{i < j} r^{\text{ell}}(z_i - z_j, \tau, \lambda) dz_{ij} - \partial_\tau(\cdot) d\tau$ is equivalent to the dynamical CYBE after this gauge transform. The bare collision residue is the non-dynamical KZB/Belavin input.

Remark 6.3 (Heat-equation prefactor). The KZB connection contains a heat-equation term ∂_τ coupled to the quadratic Cartan pairing. With the standard theta normalization, the heat equation is

$$4\pi i \partial_\tau \theta(z, \tau) = \partial_z^2 \theta(z, \tau).$$

This fixes the diagonal prefactor in the KZB connection. Off-diagonal root terms are normalized by the same theta-function convention, not by a separate numerical check.

Remark 6.4 (Normalised Szegő/Fay identity as universal modular identity). The Fay trisecant identity, originally for $g \geq 2$ theta functions, specializes at $g = 0$ and $g = 1$ to the normalised Szegő/prime-form identity [18, Cor. 2.5], which holds universally at all $g \geq 0$ after the theta-prime and prime-form

normalisations are included. This normalised identity is the modular compatibility that replaces the pentagon in the elliptic and higher genus monodromy computations; at $g = 1$ it controls the Felder dynamical associator, at $g = 2$ it underlies the doubly-dynamical DDYBE. The Szegő/Fay identity is *distinct* from the theta-nulls quartic identity on the theta-divisor (which is a $g = 2$ -specific statement).

Remark 6.5 (Higher genus). At genus $g \geq 1$, the collision residue receives corrections from the higher-genus projections of $\Theta_{\mathcal{A}}$. The Gaudin Hamiltonians generalize to commuting operators on $\overline{\mathcal{M}}_{g,n}$, and the connection ∇^{GZ} acquires curvature proportional to $\kappa \cdot \omega_g$ (the Hodge class). The genus-zero seven-face identification extends to higher genus on the uniform-weight lane; for multi-weight algebras at $g \geq 2$, the cross-channel correction $\partial F_g^{\text{cross}} \neq 0$ introduces additional terms.

7 Compute verification

Computation 7.1 (Engine catalogue). The results are verified by two compute engine suites:

- (i) *Gaudin-collision engine* (105 tests): verifies the identification $H_i^{\text{GZ}} = H_i^{\text{Gaudin}} / (k + h^\vee)$ for $\mathfrak{sl}_2$, $\mathfrak{sl}_3$, $\mathfrak{sl}_4$, $\mathfrak{so}_5$, and G_2 at multiple levels; verifies the operator-order trichotomy for all standard families; verifies the Virasoro collision residue pole structure; checks commutativity $[H_i, H_j] = 0$ numerically for $n = 3, 4, 5$ points.
- (ii) *Seven-face cross-check engine* (72 tests): verifies the seven-face chain at genus zero; checks the three-parameter identification $\hbar_{\text{KZ}} = \hbar_{\text{DNP}} = \hbar_{\text{bar}} = 1/(k + h^\vee)$; verifies the $\mathcal{W}_3$ collision depth $k_{\text{max}} = 5$ and operator order 4; cross-checks against the Yangian R -matrix and Sklyanin bracket computations.

Total: 177 tests, all passing.

Computation 7.2 ($\mathcal{W}_3$ collision residue). The $\mathcal{W}_3$ collision residue is computed explicitly. The W -self-OPE at the z^{-6} pole is

$$W(z)W(w) \sim \frac{c/3}{(z-w)^6} + \dots$$

After d log absorption: $r_{WW}(z) = (c/3)/z^5 + \dots$. The full collision residue, including the TT , TW , and WW sectors, has the structure

$$r_{\mathcal{W}_3}(z) = \frac{c/3}{z^5} + \frac{\alpha_3}{z^3} + \frac{\alpha_2}{z^2} + \frac{\alpha_1}{z}$$

where $\alpha_3, \alpha_2, \alpha_1$ are elements of $\mathcal{W}_3^! \otimes \mathcal{W}_3^!$ depending on c and the $\mathcal{W}_3$ structure constants. The z^{-2} coefficient is a lower-symbol term from composite-field reduction; it does not affect the fourth-order top symbol from z^{-5} .

References

- [1] B. Feigin, E. Frenkel, N. Reshetikhin, *Gaudin model, Bethe ansatz and critical level*, Comm. Math. Phys. **166** (1994), 27–62.
- [2] D. Gaiotto and K. Zeng, *Interface minimal model holography and topological string theory*, arXiv:2603.08783, 2026.
- [3] T. Dimofte, W. Niu, and V. Py, *Line operators in 3d holomorphic QFT: meromorphic tensor categories and dg-shifted Yangians*, arXiv:2508.11749, 2025.
- [4] A. Z. Khan and K. Zeng, *Poisson vertex algebras and three-dimensional gauge theory*, arXiv:2502.13227, 2025.

- [5] V. G. Drinfeld, *Hopf algebras and the quantum Yang-Baxter equation*, Dokl. Akad. Nauk SSSR **283** (1985), 1060–1064.
- [6] V. G. Drinfeld, *Quasi-Hopf algebras*, Leningrad Math. J. **1** (1990), 1419–1457.
- [7] P. Etingof and D. Kazhdan, *Quantization of Lie bialgebras I*, Selecta Math. (N.S.) **2** (1996), 1–41.
- [8] V. G. Kac, *Infinite-Dimensional Lie Algebras*, 3rd ed., Cambridge University Press, Cambridge, 1990.
- [9] M. A. Semenov-Tian-Shansky, *What is a classical r -matrix?*, Funct. Anal. Appl. **17** (1983), 259–272.
- [10] M. Gaudin, *Diagonalisation d’une classe d’hamiltoniens de spin*, J. Physique **37** (1976), 1087–1098.
- [11] B. Feigin, E. Frenkel, *Affine Kac–Moody algebras at the critical level and Gelfand–Dikii algebras*, Int. J. Mod. Phys. A **7**, Suppl. 1A (1992), 197–215.
- [12] D. Bernard, *On the Wess–Zumino–Witten models on the torus*, Nucl. Phys. B **303** (1988), 77–93.
- [13] A. A. Belavin, *Dynamical symmetry of integrable quantum systems*, Nucl. Phys. B **180** (1981), 189–200.
- [14] A. A. Belavin and V. G. Drinfeld, *Solutions of the classical Yang–Baxter equation for simple Lie algebras*, Funct. Anal. Appl. **16** (1982), 159–180.
- [15] G. Felder, *Elliptic quantum groups*, in *Proceedings of the International Congress of Mathematicians*, Zürich, 1994.
- [16] G. Felder and A. Varchenko, *On representations of the elliptic quantum group $E_{\tau, \eta}(\mathfrak{sl}_2)$* , Comm. Math. Phys. **181** (1996), 741–761.
- [17] P. Etingof and A. Varchenko, *Solutions of the quantum dynamical Yang–Baxter equation and dynamical quantum groups*, Comm. Math. Phys. **196** (1998), 591–640.
- [18] J. D. Fay, *Theta Functions on Riemann Surfaces*, Lecture Notes in Mathematics, vol. 352, Springer, Berlin, 1973.
- [19] R. Lorgat, *Modular Koszul duality: a monograph on bar-cobar duality for chiral algebras on algebraic curves*, Volume I (manuscript, 2026).
- [20] R. Lorgat, *A -infinity chiral algebras and 3d holomorphic-topological QFT*, Volume II (manuscript, 2026).